

Business Question and Visualization Report

Date	20 Sep 2026
Team ID	Master of Computer Applications
Project Name	Online Learning Engagement and Student Performance Analysis
Maximum Marks	5 Marks

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting

Business Questions and Visualisation

The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations in Power BI. Well-framed questions help in identifying key metrics, selecting relevant data, and building visualisation that provide insights.

Sample

Question 1: Overall Final Result Distribution

- **Business Question:** What is the distribution of students across different final academic results (Pass, Fail, Distinction, Withdrawn)?
- **Visualization: Pie Chart / Donut Chart** showing the count and proportion of students for each final result status.

Question 2: Student Academic Performance Across Regions

- **Business Question:** How does student academic performance (average score) vary across different geographical regions?
- **Visualization: Clustered Bar Chart / Map Visualisation** comparing the average student assessment score across regions.

Question 3: Correlation Between Engagement Level and Student Final Result

- **Business Question:** How does student interaction and VLE engagement level (total clicks) influence their likelihood of passing or withdrawing?
- **Visualization: Stacked Column Chart** showing the count of students categorized by Engagement Level (Low, Medium, High) stacked by Final Result.

Question 4: Academic Performance by Prior Educational Background

- **Business Question:** Which highest education qualification category yields the highest average scores and completion rates?
- **Visualization: Clustered Column Chart** comparing average scores and pass rates across prior education levels (e.g., A Level, HE Qualification, Lower Than A Level).

Question 5: Risk Level Analysis by Socioeconomic Status (BAND IMD)

- **Business Question:** How is student academic risk level distributed across different Index of Multiple Deprivation (IMD) bands?
- **Visualization: 100% Stacked Bar Chart** showing the proportion of risk levels (Low Risk, High Risk, Very High Risk, Safe) across IMD deprivation deciles.

Question 6: Relationship Between Studied Credits and Dropout Rates

- **Business Question:** Does taking on a higher number of studied credits increase the risk or rate of student dropout?
- **Visualization: Line & Stacked Column Chart** plotting the total number of students enrolled per credit load against the dropout rate percentage.

Question 7: Gender-Based Comparison of Performance and Risk Levels

- **Business Question:** Are there significant differences in average scores and risk levels between male and female students?
- **Visualization: Matrix / Clustered Bar Chart** comparing total clicks, average scores, and risk distributions split by gender.

Question 8: Total Platform Interaction vs. Academic Achievement

- **Business Question:** What is the relationship between overall student VLE activity (total clicks) and final performance level?
- **Visualization: Scatter Plot** displaying total clicks on the X-axis against average score on the Y-axis, segmented by performance level (High, Medium, Moderate).